Amrutvahini College of Engineering, Sangamner

Department of Electronics and Telecommunication Engineering

Experiment No.: 10

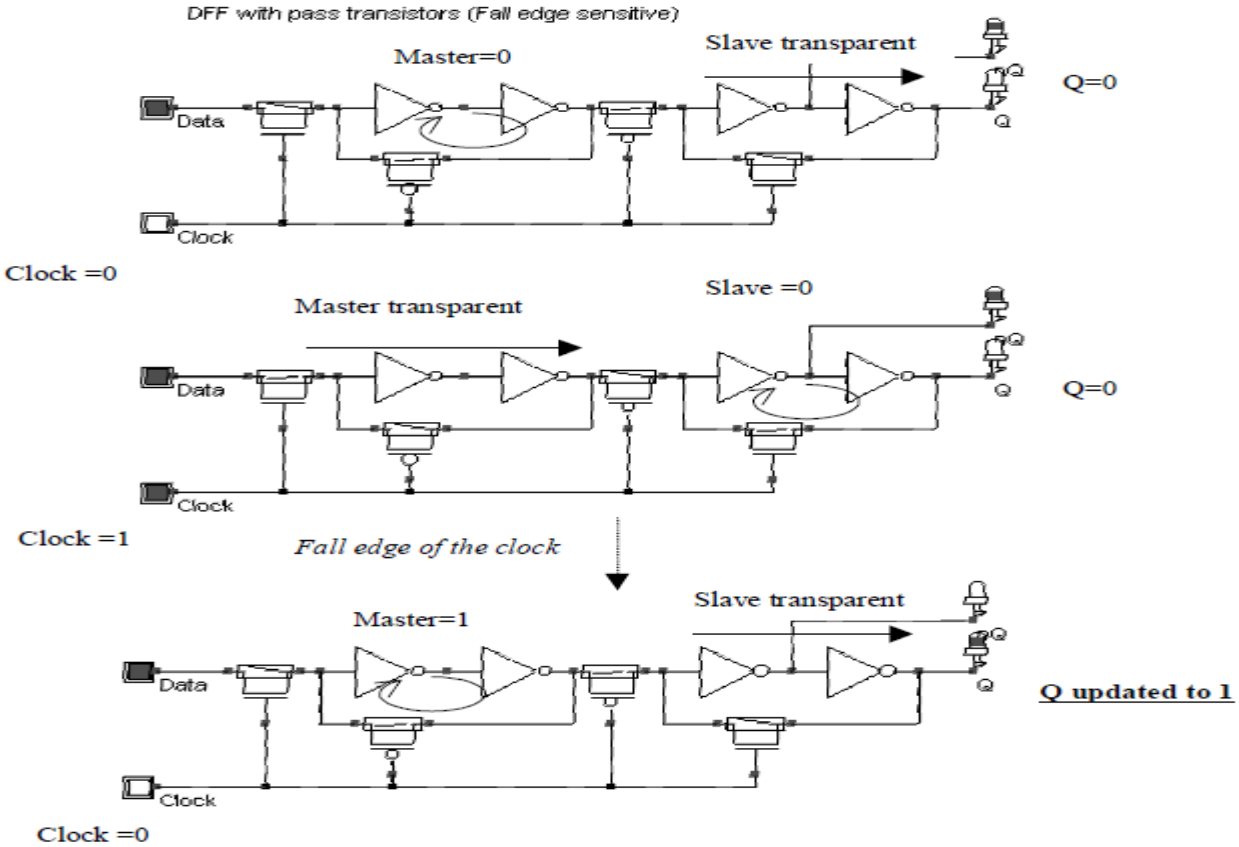
Date of Performance:

AIM: - To design D – flip flop layout.

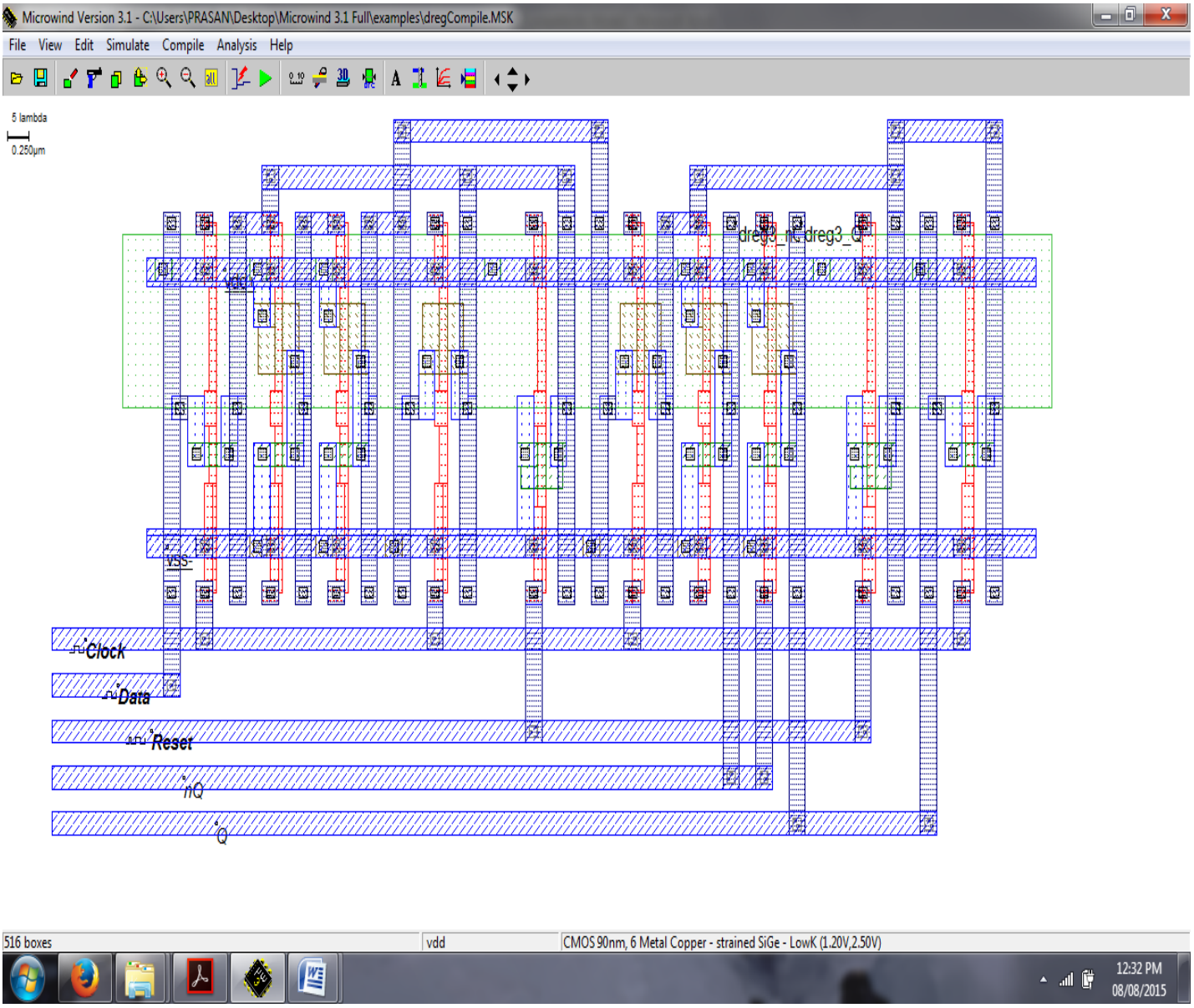
OBJECTIVE: - To study design of D – flip flop layout in Microwind and verify the result.

THEORY: -

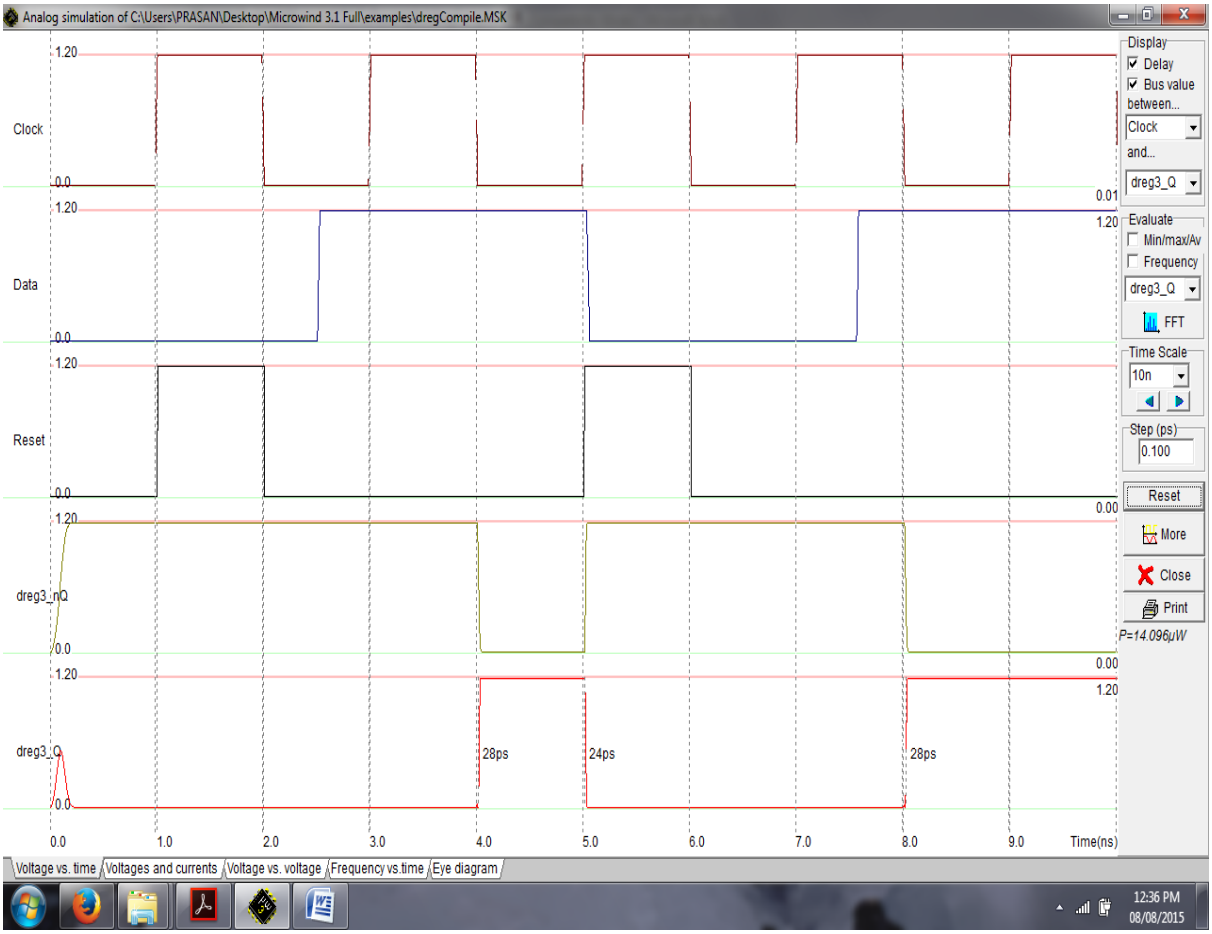
This edge-trigged latch is one of the most widely used cells in microelectronics circuit design. The cell structure comprises two master-slave basic memory stages. The most compact implementation of the edge-trigged latch is reported below. The schematic diagram is based on inverters and pass-transistors. On the left side, the two chained inverter are in memory state when the pMOS loop transistor *P1* is on, that is when *Clk=0*. The two-chained inverters on the right side act in an opposite way. The reset function is obtained by a direct ground connection of the master and slave memories, using nMOS devices. When *clock* is high, the master latch is updated to a new value of the input *D*. The slave latch produces to the output *Q* the previous value of *D*. When *clock* goes down, the master latch turns to memory state. The slave circuit is updated. The change of the clock from 1 to 0 is the active edge of the clock. This type of latch is a negative edge flip flop.



Schematic:


Microwind Version 3.1 - C:\Users\PRASAN\Desktop\Microwind 3.1 Full\examples\dregCompile.MSK
File View Edit Simulate Compile Analysis Help
5 lambda
0.250µm
dreg3_nQ
dreg3_Q
vdd
vss
Clock
Data
Reset
nQ
Q
516 boxes
vdd
CMOS 90nm, 6 Metal Copper - strained SiGe - LowK (1.20V,2.50V)
12:32 PM
08/08/2015

Waveform:



Conclusion:-

Questions:

1. Explain Fabrication methods of circuit element in details.
2. Write short note on 1. Layout of Cell 2. Library cell designing for NAND & NOR.
3. Explain different types of Fault in detail.
4. What is built in Self-test (BIST). Explain in detail.

Reference Book:

Neil H. E. West, David Money Harris, "CMOS VLSI Design(A Circuits and System perspective)", Fourth Edition, Pearson Education, Inc., publishing as Addison-Wesley, ch.1, pp-12-15.